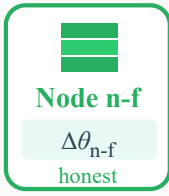


Byzantine-Resilient Consensus Protocol

DISTRIBUTED EDGE NODES



...



$\{\Delta\theta_i\}_{i=1}^n$

PHASE 1 Trust Scoring



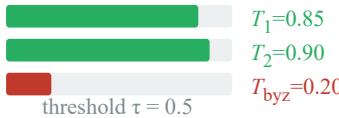
Trust Score Update

$$T_i^{(t)} = \alpha T_i^{(t-1)} + (1-\alpha)\rho_i$$

$$\rho_i = \cos(\Delta\theta_i, \Delta\theta_{\text{median}})$$

$\alpha = 0.9$ (momentum)

Trust Distribution



PHASE 2 Statistical Filtering



Multi-Krum

Select k updates with smallest neighbor dist.

$$k = n - f - 2 \quad \checkmark$$



Coord. Median

Robust aggregation per coordinate

$$\text{median}_i(\Delta\theta_i[j]) \quad \checkmark$$



Norm Bound

Clip gradient norm to bound τ

$$\|\Delta\theta_i\| \leq \tau \quad \checkmark$$

filtered $\{\Delta\theta_i\}$

PHASE 3 Multi-Strategy Aggregation



Weighted Aggregation

$$\theta_{\text{global}}^{(t+1)} = \theta_{\text{global}}^{(t)} + \sum_{i \in S} w_i \Delta\theta_i$$

$$w_i = T_i |D_i| / \sum_j T_j |D_j|, S = \text{selected honest}$$



Global Model

θ_{global}

broadcast to all nodes

PBFT Guarantees



Safety

Honest nodes agree on same global model



Liveness

Protocol terminates within bounded time



BFT Bound

$$f < n/3$$

Convergence Properties



Rate: $O(1/\sqrt{T})$

Comm: $O(nd)/\text{round}$

Rounds: T epochs

Legend: Honest Byzantine Trust Filter Aggregate PBFT